

NEW INITIATIVES IN EARTHQUAKE MANAGEMENT IN INDIA

In the recent past, we have witnessed several devastating earthquakes in different parts of the world, including a few developing countries as well as some developed countries. The 7.9 magnitude Sichuan earthquake in Southwest China on 12th May 2008 resulted in more than 5 million people becoming homeless and it became a grim reminder of the devastating Tangshan earthquake of 1976 in northeastern China which killed over 300,000 people and destroyed several million houses. The Sichuan earthquake, seen as China's worst earthquake in more than 30 years, claimed 69,195 lives and caused injury to 374,176 people, with 18,389 people reported as missing. In Pakistan, the Muzaffarabad earthquake of 8th October 2005 killed more than 88,000 people, including several thousand school children who lost their lives due to the collapse of school buildings. Even in India, several earthquakes have occurred in the recent past, which resulted in death and injury to large number of people due to the collapse of buildings, apart from the crippling impact of economic losses caused by the damage to property, assets and infrastructure.

We have often heard this statement: "Earthquakes don't kill; the collapse of buildings do." In the developing countries, weak compliance of building codes and town planning byelaws and inadequate attention to enforcement and regulation are contributing to the loss of lives, injuries and loss of property when earthquakes strike. In the developed world, even a high intensity earthquake causes minimal loss of lives and property as the construction of buildings, public infrastructure and amenities in these countries is governed by strict regulations which ensure public safety. Thus, a high intensity earthquake in the United States of America or Japan may not lead to the loss of lives of a large number of people because the compliance with a rigorous techno-legal regime will ensure that few buildings will collapse resulting in low loss of lives. Even in the developing countries, recent earthquakes have shown that while the buildings constructed in strict compliance with earthquake-resistant building codes and other safety regulations have been able to withstand the impact of the earthquakes, poorly constructed buildings in the same neighbourhood have collapsed resulting in loss of lives and destruction of property and assets.

The Muzaffarabad earthquake in Pakistan and the Sichuan earthquake in China have reiterated the critical need to comply with earthquake-resistant building codes and other safety regulations in earthquake-prone areas. One of the most important lessons from these earthquakes is the critical imperative of ensuring the structural integrity of public assets like schools and hospitals and private assets like houses, commercial buildings, etc. to withstand the intensity of devastating earthquakes. Thus, the proactive action of all stakeholders

will minimize the loss of lives and damage to assets and infrastructure in earthquake-prone areas and we need to create greater public awareness on the earthquake risk and vulnerability in India.

Earthquake Risk and Vulnerability in India

India's high seismic risk and vulnerability is evident from the fact that 10.9% of its geographical area falls in the seismic zone V which is vulnerable to very high seismic risk, while 17.3% of its geographical area falls in seismic zone IV vulnerable to high seismic risk. According to the Vulnerability Atlas of India prepared by the Building Materials Technology Promotion Council (BMTPC), 229 districts of India fall within seismic zones IV and V. This Vulnerability Atlas has been prepared based on the past trends of earthquakes in the country and efforts are now under way to prepare a probabilistic seismic hazard map for India, with the help of seismic experts and scientific institutions, to more realistically reflect the seismic risk and vulnerability profile of the country.

The recorded history of earthquakes in India indicate that India has faced four very high intensity earthquakes of magnitude 8.0 or higher since 1890. The relatively low damage and devastation caused by these high intensity earthquakes faced by India in Shillong in 1897 (M 8.7), Kangra in 1905 (M 8.0), Bihar-Nepal in 1934 (M 8.3) and Assam-Tibet in 1950 (M 8.6) was largely because of the types of construction and building materials used in such places. It is estimated that the occurrence of such a very high intensity earthquake now in any of the highly densely populated regions would result in very high loss of lives and injuries to large numbers of people, beyond the coping capacity of

the emergency medical services facilities and personnel in these areas. With the rapid spread of urbanisation and economic development, trends in housing construction in most urban, suburban and rural areas indicate sudden increases in reinforced cement concrete (RCC) construction. In the areas of high seismic risk, these result in collapse of or damage to such structures thereby causing loss of lives and cause injury to people.

Seismic Zone Map of India (IS : 1893, 2002)

